

SFANTU GHEORGHE, Romania

WMO: 152380

Lat: 45.8717N

Lon: 25.8022E

Elev: 524

StdP: 95.18

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-14.0	-18.8	0.8	-16.7	-16.0	1.0	-13.2	9.2	-0.1	7.4	-2.4	0.4	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	31.0	19.5	29.4	19.0	27.8	18.6	21.0	28.2	20.2	27.1	19.4	26.0	1.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.4	23.3	17.9	13.7	22.4	17.2	13.1	21.6	63.1	28.2	60.2	27.0	57.6	26.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.6	-0.4	3.5	8.6	14.2	18.1	19.2	19.4	14.6	9.0	4.2	-0.9
	DBStd	8.85	4.92	5.06	4.31	4.63	3.11	2.92	2.63	2.74	4.01	3.58	4.08	4.44
	HDD10.0	1592	422	291	205	80	6	1	0	0	8	62	178	338
	HDD18.3	3604	680	524	461	293	130	38	23	22	122	289	424	597
	CDD10.0	1178	0	0	2	38	137	244	284	292	145	31	4	0
	CDD18.3	149	0	0	0	0	3	32	49	56	9	0	0	0
	CDH23.3	2114	0	0	0	24	104	433	590	752	203	7	0	0
	CDH26.7	622	0	0	0	1	13	113	184	262	50	0	0	0
(14) Wind	WSAvg	1.3	1.2	1.2	1.6	1.6	1.4	1.3	1.4	1.2	1.3	1.1	1.0	1.1
(15) Precipitation	PrecAvg	585	23	22	28	51	77	90	83	64	50	42	28	28
	PrecMax	848	54	36	52	92	172	152	152	119	102	97	59	67
	PrecMin	423	4	6	9	19	32	23	35	24	7	2	1	5
	PrecStd	100	11	8	11	22	32	38	35	30	24	24	15	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	15.9	20.2	25.6	28.3	31.6	32.4	33.2	30.4	24.3	18.6	10.7
		MCWB	6.7	10.2	10.7	14.0	16.3	20.8	20.2	20.3	18.2	15.1	12.2	7.9
	2%	DB	6.1	11.7	16.4	23.5	26.1	29.4	30.6	31.1	27.8	21.8	16.0	8.0
		MCWB	4.2	7.5	9.1	13.3	16.0	19.7	19.6	19.2	17.3	14.0	11.2	6.1
	5%	DB	3.9	9.1	13.4	20.9	24.1	27.7	28.7	29.5	25.6	19.4	13.3	5.9
		MCWB	2.5	5.8	7.5	12.4	15.7	19.2	19.3	18.8	16.7	13.1	9.8	4.4
	10%	DB	2.4	6.1	10.8	17.8	22.1	25.7	26.7	27.6	23.4	16.3	11.0	4.1
		MCWB	1.3	3.8	6.2	11.1	14.6	18.1	18.7	18.6	16.0	11.9	8.1	3.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	10.5	11.5	15.2	18.3	22.3	21.6	21.8	19.5	16.0	13.0	8.5
		MCDB	8.9	15.7	19.8	24.0	24.9	29.6	28.5	30.4	28.0	21.7	17.6	10.2
	2%	WB	4.4	7.8	9.7	14.0	17.1	20.7	20.8	20.8	18.2	14.7	11.7	6.1
		MCDB	5.9	11.7	16.0	21.5	23.8	27.7	27.7	28.4	25.6	20.6	15.3	8.0
	5%	WB	2.9	6.0	8.2	12.9	16.2	19.7	20.0	19.9	17.3	13.6	10.4	4.6
		MCDB	3.9	8.9	13.2	20.2	22.7	26.4	26.5	27.2	24.0	18.3	13.7	6.0
	10%	WB	1.5	4.2	6.8	11.6	15.3	18.7	19.3	19.1	16.4	12.3	8.6	3.2
		MCDB	2.4	5.9	10.3	17.1	21.1	24.6	25.5	25.9	22.6	16.0	11.1	4.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	10.0	11.0	12.7	13.0	13.2	13.2	14.9	13.4	12.6	9.9	6.7
		MCDBR	8.8	13.6	17.2	19.3	17.4	16.6	17.2	18.3	18.3	19.3	15.0	9.1
		MCWBR	6.9	9.2	10.0	9.6	8.2	7.4	7.0	7.2	8.2	11.3	9.8	7.2
	5% WB	MCDBR	8.4	13.5	16.1	17.2	15.6	15.3	14.8	16.1	16.5	17.3	14.4	8.7
		MCWBR	6.7	9.4	9.5	8.8	7.6	7.3	6.5	6.8	7.5	10.4	9.4	6.7
(40) Clear-Sky Solar Irradiance	taub		0.303	0.332	0.376	0.435	0.414	0.412	0.430	0.442	0.397	0.359	0.332	0.304
	taud		2.335	2.266	2.223	2.100	2.232	2.272	2.228	2.189	2.285	2.381	2.405	2.372
	Ebn at Noon		791	833	844	821	855	856	834	806	813	796	752	749
	Edh at Noon		76	100	121	149	135	131	135	135	113	89	72	67
(44) All-Sky Solar Radiation	RadAvg		1.34	2.16	3.27	4.34	5.37	5.77	5.83	5.30	3.84	2.60	1.48	1.08
	RadStd		0.15	0.26	0.35	0.45	0.49	0.50	0.49	0.43	0.39	0.32	0.16	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (70 neighbors)	+0.62	N/A	N/A	+0.69	N/A	N/A	-138	-183	+77	+24

Nomenclature: See separate page